Biomedical Engineering Advances 11 (2026) 100215

Contents lists available at ScienceDirect

Biomedical Engineering Advances

journal homepage: www.journals.elsevier.com/biomedical-engineering-advances

Artificial intelligence for epileptic seizure prediction: Toward hybrid, patient-centric, and edge-ready systems

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ARTICLE INFO

Keywords:
Epilepsy
Seizure prediction
Deep learning
Personalization
Time-series analysis

ABSTRACT

This review provides a comprehensive examination of artificial intelligence (AI) approaches for epileptic seizure prediction, emphasizing engineering solutions that enhance accuracy, scalability, and clinical applicability. While generalized models, often trained on large multi-patient datasets, have advanced the field, they remain limited in capturing the neurological variability required for reliable patient-specific predictions. This work analyzes recent progress in processing electroencephalography (EEG), electrocardiography (ECG), and electromyography (EMG) signals, and evaluates cutting-edge deep learning methods for feature engineering, classification, and deployment in resource-constrained environments. Key challenges in current systems are identified, including limited availability of patient-specific data, a lack of standardized evaluation practices, and high computational demands that hinder real-time use. The primary AI contribution of this paper is the proposal of a novel hybrid prediction framework. This framework integrates a generalized deep learning backbone for robust feature representation with a lightweight personalization layer, designed to adapt predictions using minimal individual data through techniques like transfer learning and Dynamic Time Warping. The engineering application focuses on enabling this framework for practical, real-world adoption by incorporating feature reduction, model compression, and hardware-aware optimization. This allows for deployment on wearable and edge-computing platforms, ensuring computational efficiency and sustainability. By combining robust generalization with essential personalization, this paper establishes a blueprint for the next generation of sustainable Artificial Intelligence-driven seizure prediction systems and highlights engineering-driven pathways toward clinically viable neurotechnology.

1. Introduction

1.1. Neurological challenges

The quality of life for elderly individuals with neurological disorders is a complex issue, encompassing challenges in cognitive function, motor abilities, and emotional well-being. Conditions like Dementia or Parkinson's disease can erode independence, leading to increased reliance on caregivers [1–3]. Routine activities, social interactions, and overall functioning are often compromised. Supportive care, appropriate medical interventions, and initiatives promoting mental and physical well-being are crucial in improving the overall quality of life for elderly individuals facing neurological challenges [2].

Epilepsy, known for its irregular and repetitive seizures, can be described as a neurological disorders that affects over 50 million people in the world [3–7]. Epileptic seizures are best described as sudden discharge and abnormality in the electrical activity of the brain [6]. The reoccurring epileptic seizures can change the development of the nervous system by adjusting the receptor expression and distribution, even if there is no damage to the neurons. This can result in modifications to behavior and cognitive function [8].

Currently, Anti-Epileptic Drugs (AEDs) can effectively manage the condition of seizures for approximately 70 % of the patients. Nonetheless, it is noteworthy that the medications prove to be ineffective for a significant demographic comprising approximately 15 million individuals who suffer from epilepsy [3,9–11].

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https://doi.org/10.1016/j.bea.2026.100215
Received 11 October 2025; Received in revised form 4 March 2026; Accepted 4 March 2026
Available online 6 March 2026

Further, convulsive seizures, if neglected, may lead to partial physical disability and in some cases even death [12]. Understanding these seizure types is crucial for accurate diagnosis and tailored treatment. Research continues to explore novel therapeutic interventions and a deeper understanding of the underlying mechanisms of epilepsy. This paper investigates the effectiveness of personalized seizure predictors against generalized models, exploring whether personal factors such as age and gender can enhance prediction accuracy, and comparing various optimization techniques used to improve the efficiency and speed of deep learning models in seizure prediction. This pursuit aims to enhance the efficacy of existing treatments and develop new strategies to address the challenges faced by those who do not respond to current medications.

1.2. Epileptic seizures

Although epilepsy is found in all ages and genders, the experience of the disease varies from one patient to another [6]. These episodes occur in various forms, each characterized by distinct clinical features and underlying neurological mechanisms. The two main categories of epileptic seizures are focal (partial) seizures and generalized seizures [13] as shown in Fig. 1.

1. **Partial Seizures (Focal)** only affect a section of the cerebral hemisphere. Partial seizures are further classified depending on the impact they have on the patient. If the symptoms are mild and the patient remains conscious, it is referred to as simple-partial. However, seizures that cause confusion and abnormal behaviour in patients are categorized as complex partial [6,13].
2. **Generalized Partial** seizures quickly affect all sections of the brain and are often further categorized to Absence and Tonic-Clonic seizures [6,13]. Absence Seizures are characterized by a sudden loss of consciousness and responsiveness, often accompanied by subtle movements like eye blinking or lip smacking. Tonic-Clonic Seizures Formerly known as grand mal seizures, these involve loss of consciousness, stiffening (tonic phase), followed by rhythmic jerking of the limbs (clonic phase) [14].

1.3. Seizure detection and prediction

Although real-time seizure detection can lead to quick response form the caretakers, it heavily relies on physically attentive caretakers at all-time. Unpredictability of epileptic seizures can severely impact the quality of life of patients, leaving them with uncertainty at any time of the day [16]. Forecasting of seizures can drastically enhance the control over the mental disorder, thus, improving patient's life [16,17].

Recent studies by several neurologists indicate the presence of a state where the risk of seizure is heightened [18]. Consequently, the detection of pre-ictal state can predict the seizure up top 45 min prior to occurrence. In the prediction field, EEG is a dominant data type used across all neuro activity before and after the seizure [17,18].

1.4. Paper objective and structure

Existing survey papers concentrate on the variety of machine and deep learning algorithms for pre-processing, classification and prediction of epileptic seizures [6,19–23]. In contrast, this paper is dedicated to providing a comprehensive overview of recent developments specifically in epileptic seizure prediction. It further reviews the current landscape of publicly available datasets and highlights the role of biomedical signals in seizure detection and prediction tasks. While the existing literature addresses seizure detection, classification, and prediction in a broad sense, most prior reviews place a stronger emphasis on detection and classification. As summarized in Table 1, seizure prediction particularly from a personalized modeling perspective remains comparatively underexplored. This review therefore focuses on individualized seizure prediction strategies rather than generalized models, offering a distinct and timely perspective. Moreover, as indicated in Table 1, prior surveys provide limited discussion on the computational efficiency of artificial intelligence models. By examining input selection strategies and model minimization techniques, this paper addresses this gap and contributes a focused discussion on optimizing AI models for sustainable and personalized seizure prediction.

Section 2 provides an overview of the background and biomedical signal modalities used in seizure prediction. Section 3 describes various seizure prediction frameworks, including data acquisition and classification methodologies. Section 4 discusses the contrast between personalized and generalized models. Section 5 evaluates optimization techniques for improving computational and environmental efficiency. Finally, Section 6 highlights research gaps, analyzes key barriers to clinical deployment, and introduces a proposed hybrid seizure prediction framework that integrates generalized and personalized modeling approaches for improved accuracy and real-world applicability.

2. Background

2.1. Bio-medical signals

The majority of the work in the prediction field is done on the basis of electroencephalography (EEG) signals [16–18,24]. EEG (Electroencephalography) signals are electrical impulses recorded from the brain's surface, generated by the synchronized activity of neurons [25]. These signals reflect the brain's electrical patterns, which vary depending on the state of consciousness, mental activity, and neurological conditions [25]. Epileptic seizures are the result of surge in brain's electric activity [3]. Electroencephalography (EEG) provides sophisticated signals extracted from the brain's electrical activity [6,26]. Manual detection of seizures from EEG signals can be a challenging task that requires medical experts to analyse the readings one-by-one. EEG signal contains a significant amount of information related to the occurrence of seizures in both time and frequency domain [6]. Furthermore, video electroencephalography (VEEG) monitoring has recently gained popularity in seizure diagnosis and early detection as well [27–29]. The VEEG is a recording of the patients' physical state alongside the normal EEG recording from the head. The advantage of VEEG is that it allows identification of seizures and their impact on the motor abilities of the patients [29].

Certain types of seizures cause a significant increase in the patient's

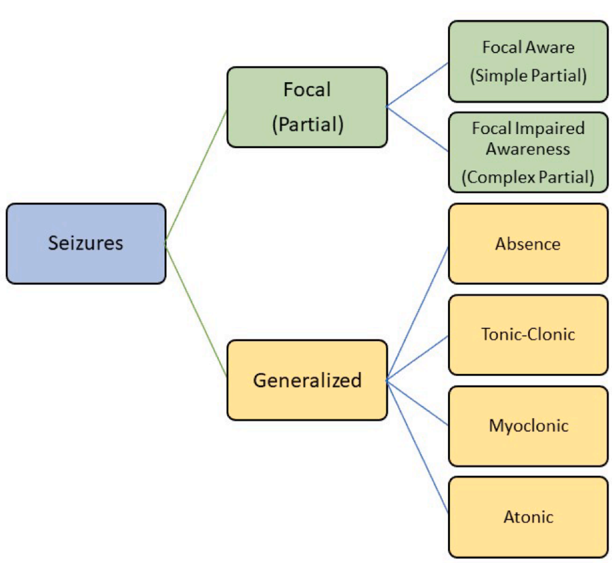

Fig. 1. Epileptic seizure types [15].

Table 1
Comparison of This Review with Recent Survey Papers on Epileptic Seizure Prediction.

Review Paper	Focus Area	Personalized Prediction	Generalized vs Personalized Comparison	Optimization / Efficiency	Edge / Real-Time Consideration
[19]	Detection & prediction methods	X	X	X	X
[6]	DL for EEG seizure analysis	Limited	X	X	Limited
[23]	Feature extraction & ML	X	X	X	X
[20]	ML/DL survey	Limited	X	X	X
[22]	cross-patient approaches	Yes	Yes	Limited	X
This work	**Prediction-focused, patient-centric**	**Yes**	**Yes**	**Yes**	**Yes**

heart rate [24]. Thus, close monitoring of heart-based bio-signals such as Electrocardiography (ECG) and Heart Rate Variability (HRV) can be beneficial in seizure detection [30,24]. Similar to EEG, ECG is a recording of electrical activity of the heart that includes the rhythm, electrical impulses and timing of contractions [31]. HRV is derivative measure from the ECG that indicates variations in time intervals between consecutive heartbeats. It reflects the autonomic nervous system's regulation of the heart, specifically the balance between the sympathetic (fight or flight) and parasympathetic (rest and digest) responses [32]. Further, recent studies by Jeppesen et al. [30] indicates that utilizing HRV bio-signals, the detectors can distinguish between physical exercise and seizures. Consequently reducing the false positive and enhancing the system sensitivity [30].

EEG and ECG approaches investigate and analyse the changes and the behaviour of internal organs. However, alternative methods look at the effects of the seizures that can be visually observed. For instance, generalized onset motor seizures can cause sudden uncontrollable body movements [24]. In these cases, measuring the movement of limbs through Electromyography (EMG) sensors could be an indication of presence of seizure in epileptic patients [24]. EMG-enabled wearables are popular solutions for detecting such seizures with high sensitivity [30]. However, considerable number of seizures do not cause random body movements, therefore the EMG-based detectors will fail to capture them. Similarly, vision-based monitoring will struggle with such types of seizures.

In summary, EEG signals are observed to be the predominant source of information in epileptic seizure detection. However, collecting EEG data can be a challenging and uncomfortable experience for the patients [3]. Alternative biomedical data such as ECG and EMG are often used where comfort outweighs the accuracy in detection systems. EEG-based detectors could potentially benefit from the change in the heart rate and motion capture data. Implementation of a novel sensor-fusion unit that optimizes the performance while maintaining a certain level of comfort for the patients can substantially enhance the quality of life of people with epilepsy. To illustrate the research landscape, Fig. 2 summarizes the distribution of studies utilizing different biomedical signals for epileptic seizure prediction between 2020 and 2025. As shown in the figure, EEG-based approaches dominate the literature, accounting for the majority of published work. In contrast, ECG, iEEG, EMG, and video-based methods remain comparatively less explored, while the growing proportion of multi-modal studies indicates an increasing research interest in combining complementary biosignals to improve prediction robustness and clinical relevance.

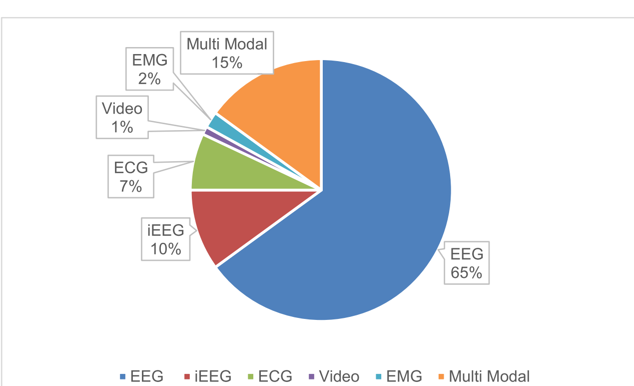

Fig. 2. Estimated Research Focus by Data Type in Epileptic Seizure Prediction (2020–2025).

2.2. *Resource analysis*

This section evaluates the resource demands of deep learning approaches for epileptic seizure prediction, focusing on the timing and computational requirements of the models. These factors are critical for assessing the feasibility of deploying such models in real-time clinical settings, where low latency and efficient resource utilization are essential.

2.2.1. *Timing analysis*

Timing analysis stands as a critical component in real-time systems, particularly within the context of health monitoring [33,34]. These systems are meticulously designed to swiftly respond to events and inputs within stringent time constraints, ensuring functionality and adherence to performance standards. The urgency for timely and accurate information in health monitoring accentuates the significance of timing analysis. It plays a pivotal role in continuous data collection from patient sensors, facilitating immediate analysis for prompt identification and response to critical health events, thereby safeguarding patient well-being. Recent research has further emphasized latency-aware algorithm design, including probabilistic prediction strategies that aim to shorten detection delay while maintaining sensitivity [35,36]. Such approaches highlight the inherent trade-off between rapid detection and false alarm control. Precision in data processing, efficient resource management, and compliance with regulatory standards further emphasize the indispensable nature of timing analysis. From closed-loop systems to timely alerts and notifications, the comprehensive integration of timing analysis ensures the reliability and quality of health monitoring systems, making it an essential consideration in their design and implementation. There are several timing parameters in seizure prediction that need to be closely monitored for real-time performance including:

A. Detection delay

Detection delay is a temporal metric used in real-time seizure detection works. This term refers to the time difference between the start of the seizure from perspective of the patient and the classifier as shown in Fig. 3.

The detection delay is also significantly influenced by factors such as batch (data packet) size, type of classification and sampling frequency. In the example shown in Fig. 3, the original seizure starts at sample number 800 while the predicted seizure is at 1150. Given the sampling frequency of 256 Hz, this error translates to 350 samples that is 1.36 s.

B. Prediction period

Similar to detection delay, prediction period represents the time

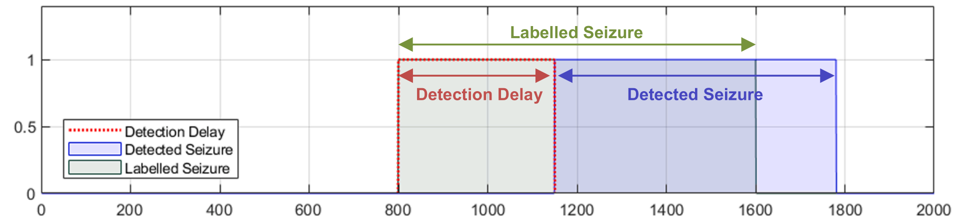

Fig. 3. Detection delay.

(often in minutes) prior to an episode of ictal state. An illustration of this metric is shown on Fig. 4. This parameter is directly proportional to the batch size and duration of sample taken into one segment of data. Numerous papers have used 5, 10, 15, 30 and 45 min segments of data.

C. Pre-processing time

Preprocessing time is the duration spent cleaning, transforming, and preparing raw data for input into a machine learning model. This involves removing noise or errors, converting the data into a usable format (such as normalizing or scaling values), and extracting key features that the neural network will use. This step is essential as the quality and format of the data greatly influence the model's performance. Frequency-domain feature extraction methods tend to produce a larger preprocessing time as compared to time-domain features.

D. Classification time/rate

Classification time is the duration it takes for a machine learning model to process a single sample and produce a classification result. This encompasses the time from when the input sample is received by the model to the moment a prediction is generated. It is a critical metric for assessing the speed and efficiency of a system, particularly in real-time or time-sensitive seizure prediction applications. Notably, classification time is influenced by several factors, including the complexity of the model architecture, the target hardware, and the amount of data included in a single input sample. Larger input windows such as multi-second EEG segments or high-resolution spectrograms increase the computational load, which in turn prolongs classification time. Experimental findings demonstrate that Graphical Processing unit (GPU)-enabled devices substantially reduce inference time compared to Computer Processing Unit (CPU-based) systems, highlighting the importance of hardware selection in time-critical implementations.

In conclusion, temporal analysis is central to the design and implementation of real-time health monitoring systems. Its impact on patient safety, data accuracy, and the overall effectiveness of healthcare interventions underscores its importance. Careful structuring of period reviews of components of health monitoring systems strengthens their reliability, ensuring that the complex requirements of the health care system are met.

2.2.2. Computing resources

Seizure prediction AI models, particularly those based on deep learning, require substantial computing resources due to the complexity and volume of the data they process. The training phase of these models involves handling large datasets of neural activity, often recorded at high frequencies, which necessitates significant computational power for data pre-processing, feature extraction, and model training. High-performance GPUs are typically used to accelerate these processes, given their ability to handle parallel computations efficiently [37]. Furthermore, real-time seizure prediction models demand not only high computational power during training but also during inference to ensure timely and accurate predictions. The need for considerable memory and storage capacities is also critical, as these models often store vast amounts of historical data and model parameters [38]. As a result, the deployment of seizure prediction models in clinical settings or wearable devices poses additional challenges in optimizing computational efficiency and ensuring the models can run on limited-resource environments without compromising performance [39]. As summarized in Table 2, seizure prediction models exhibit substantial variation in computational requirements depending on architectural complexity and input dimensionality. While traditional machine learning and shallow neural networks are well suited for real-time and edge deployment, deep learning models often demand significantly higher computational and memory resources. These differences underscore the importance of considering computing constraints alongside prediction performance, particularly for wearable and on-device seizure prediction systems.

3. Seizure prediction frameworks

3.1. Datasets

A comprehensive collection and compilation of various data types,

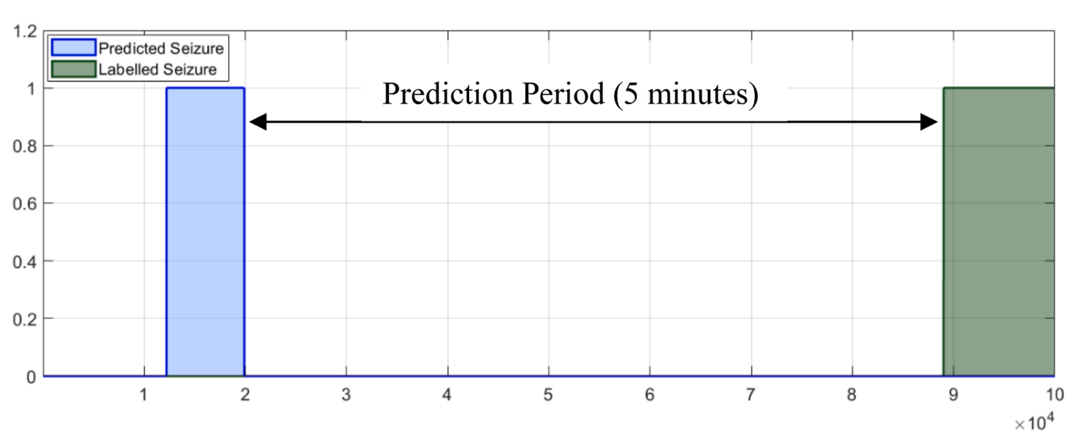

Fig. 4. Prediction Period.

Table 2
Computational Resource Characteristics of Seizure Prediction Models.

Model Category	Typical Architectures	Training Cost	Inference Cost	Memory Footprint	Representative References
Traditional ML	SVM, k-NN, Random Forest, LDA	Low	Low	Low	Acharya et al., 2018; Siddiqui et al., 2020
Shallow Neural Networks	MLP, shallow CNN	Low-Medium	Low-Medium	Medium	Ramgopal et al., 2014; Natu et al., 2022
CNN-based DL	AlexNet, ResNet, custom CNNs	High	Medium-High	High	Shoeibi et al., 2021; Truong et al., 2018
RNN-based DL	LSTM, BiLSTM, GRU	High	High	High	Tsiouris et al., 2018
Transformer-based Models	EEG Transformer, ViT	Very High	High	Very High	Yang et al., 2023
Optimized / Compressed Models	Quantized CNNs, pruned networks	Medium	Low	Low-Medium	Roy et al., 2019; Shoeibi et al., 2021

including but not limited to Electroencephalography (EEG) and Electrocardiography (ECG), plays a vital role in the seamless integration and effective evaluation of Deep/Machine learning models. Given the complexity of the problem at hand, obtaining and curating a significant amount of biomedical signals from individuals suffering from epilepsy over an extended period of time can present significant challenges. Several publicly available datasets have played a pivotal role in advancing seizure prediction research. These include the CHB-MIT, Bonn, Freiburg, TUH, and EPILEPSIAE databases, each offering diverse signal types, sampling rates, and patient counts. A detailed summary of these datasets is provided in Table 3.

3.1.1. Bonn university EEG database

This dataset includes EEG recordings of five subjects with both healthy brain activity and epileptic seizures. It's widely used for benchmarking seizure detection algorithms [40].

3.1.2. CHB-MIT scalp EEG database

Compiled by the Children's Hospital Boston and the Massachusetts Institute of Technology, this dataset contains continuous EEG recordings from pediatric subjects with epilepsy. It includes various types of seizures and is commonly used for seizure detection and prediction research. This is the currently most used dataset across multiple research projects [41,42].

3.1.3. Freiburg seizure prediction dataset

A dataset containing long-term EEG recordings of 21 patients with epilepsy, aiming to facilitate research on seizure prediction algorithms [43].

3.1.4. Temple University Hospital EEG dataset

This dataset comprises EEG recordings from patients with epilepsy. It includes various types of seizures and is utilized for seizure detection and classification studies [44]. The continuous project has reached over 14,000 patient recordings as of February 2024.

3.1.5. European epilepsy database (EPILEPSIAE)

A repository containing multi-modal data collected from multiple European clinical centers. It includes recordings from individuals with epilepsy for research purposes. This is the continuation of the Freiburg dataset that is no longer publicly available for free. Moreover, boasting a remarkable compilation of >1000 recorded seizures derived from a cohort of 161 patients, this database undoubtedly stands as one of the most comprehensive epileptic datasets available to date [45].

3.2. Feature engineering

Raw biomedical signals contain numerous artifacts that relate to a person's current physical or mental state e.g. sleeping, stress, exercise, blinking, etc. External factors such as day-to-day activity, sleeping, stress, physical exercise and blinking produce periodic, semi-periodic and irregular noise in both EEG and ECG data [23].

3.2.1. Time domain and statistical features

EEG signal contains a significant amount of information in both time and frequency domain. Time-domain features such as statistical parameters can add valuable information, however, frequency domain features have often outperformed time-domain based classifiers [23]. Maheshwari et al. [46] proposes a novel technique for real-time seizure detection based on high spatial frequencies. In this method, the EEG signals are divided into smaller overlapping segments s, where the signal can be considered stationary. A brain network can be represented via a collection of EEG channels, set of edges and the edge weight matrix. The edge matrix, calculated using Gaussian kernel weighting function, depicts the connectivity measures between EEG channels for each segment of data. The Laplacian graph is able to measure the speed at which information moves between different EEG channels. Eigenvalue decomposition of the graph provides particular information about the spatial frequency. The creation of a Sub-band Characteristic Response Vector (CRV) involves utilizing the eigenvalues and eigenvectors of the connectivity matrix that corresponds to a specific state of the brain. In the final step of the pre-processing, Euclidean distance between two successive segments generates the distance time-series, which is used as an input to the classifier as shown in Fig. 5.

Alternative methods such as real-time implantable devices have gained popularity. The authors propose a novel cascade architecture consisting of two sequential detectors, each designed to address specific aspects of seizure detection [47]. In the first stage, a lightweight detector continuously monitors Electroencephalography (EEG) signals in real-time, using a simple algorithm based on time-domain features like line-length or area. The primary goal of this stage is to quickly identify potential seizure candidates, even at the cost of generating false

Table 3
Summary of Publicly Available Epilepsy Datasets Used in Seizure Prediction Research.

Dataset	Sampling Frequency (Hz)	Data types	Number of Seizures	Number of Patients	Limitations
Bonn University	173.61	EEG	3	5	Very small dataset; limited number of patients and seizures; not representative of diverse seizure types or demographics.
CHB-MIT	256	EEG	185	23	Pediatric patients only; mostly temporal lobe seizures; limited demographic diversity; may not generalize to adults.
Freiburg	256	EEG	88	21	Primarily intracranial EEG from presurgical epilepsy patients; may not reflect general epilepsy population; limited seizure type diversity.
Temple University	256	EEG	NA	14,000	Heterogeneous clinical recordings; missing seizure labels in some cases; variable recording conditions; potential demographic biases.
EPILEPSIAE	256–1024	EEG, ECG, EMG, EOG	1102	161	Mix of intracranial and scalp EEG; patient population mostly adults; seizure types and demographics not fully balanced; recordings vary in duration and quality.

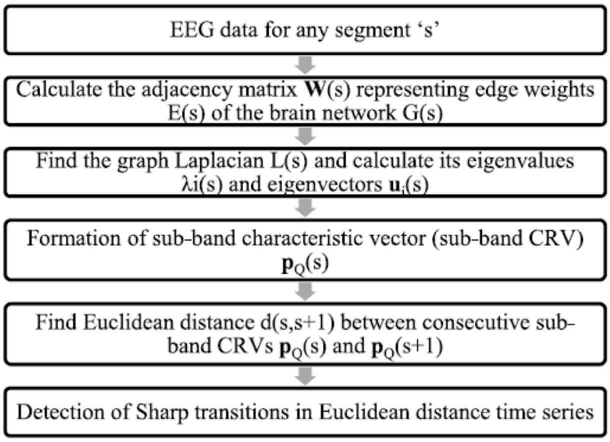

Fig. 5. High spatial frequency method.

positives [47]. The second stage comes into play only when the first detector flags a seizure candidate. This second detector employs a high-accuracy algorithm, focusing on frequency-domain features such as spectral entropy or wavelet transform. Its purpose is to carefully evaluate the identified seizure candidates from the first stage and eliminate false positives, ensuring a more reliable seizure detection [47].

3.2.2. Frequency domain features

SJ Hussein [26] investigates the effectiveness of Discrete Wavelet Transform (DWT) and Empirical Mode Decomposition (EMD) in epileptic seizure detection. In this method, the data is split into 2 s segments and labelled as seizure or non-seizure samples. The signals are decomposed via discrete wavelet transform with 5 levels of decomposition. This process produces 16 features per channel. To minimize the hardware requirement, arithmetic mean of all available channels is used as a single channel.

A new variation of wavelet named wavelet scattering transform (WST) has also gained popularity in complex-signal classification and analysis [48,49]. The WST demonstrates a favourable balance between preserving consistent features and distinguishing differences, remaining resilient to distortions, providing insightful signal representation, and preserving high-frequency details [50]. Zhang et al. [48] purposes a novel method for accurate detection of various EEG and ECG periods utilizing a framework consisting of WST, Bi-Directional PCA and grey wolf optimization. Although the target of this research is not exactly aligned with topic of this research, the methodology shows potential to for real-time seizure detection and prediction as well.

Ghassemi et al. [51] presents an accuracy-improved seizure detection algorithm utilizing combination of a variety of statistical and entropy-based features with fractal-features. The present model employs a specific type of wavelet transform known as Tunable-Q Wavelet Transform (TQWT) to perform a decomposition of brain waves as shown in Fig. 6. The resulting transformed signals are subjected to the extraction of five statistical features, namely mean, variance, standard deviation, kurtosis, and skewness, as well as seven entropy features. Additionally, Higuchi, Katz and Petrosian fractal features are appended to the set of features. The feature matrix, combining all the three main categories of features, is then separated into a training and testing set.

An alternate approach utilizes Cepstral Coefficients alongside windowing. Authors of "A Deep Learning-Based Real-time Seizure Detection System" [52], describe a system that produces 26 Linear Frequency Cepstral Coefficients (LFCC) features, including 8 absolute cepstral coefficients, a differential energy term, 9 first derivatives, and 8 s derivatives [52]. These features are derived from the EEG signal for analysis. The feature extraction module uses a 0.3-second buffer to extract 0.2-second center-aligned windows every 0.1 s from each EEG channel. Additionally, differential energy, first derivatives, and second derivatives are computed using specific window durations.

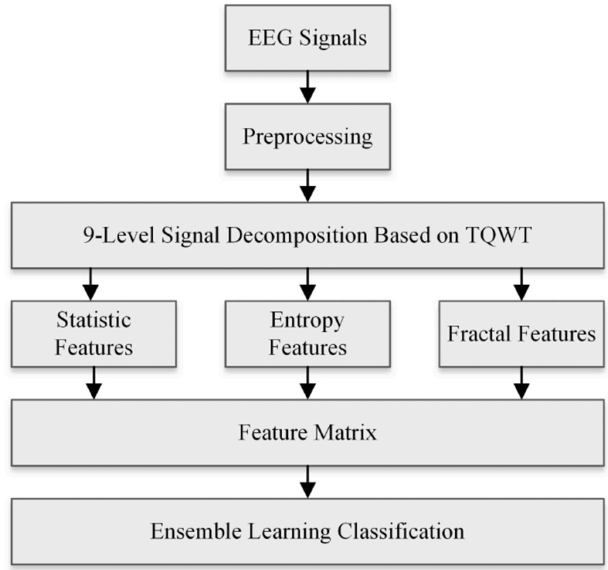

Fig. 6. Tuneable-Q Wavelet transform [51].

3.2.3. Features derived by neural networks (AI generated features)

Convolutional Neural Networks (CNNs) have revolutionized the field of feature engineering by extracting unique patterns and features from images. One prominent application of CNNs is in feature generation for datasets. Traditionally, feature engineering requires manual extraction of relevant attributes from raw data, which is labour-intensive and often limited in capturing complex patterns. CNNs, however, automate this process by learning hierarchical representations directly from the data. By utilizing multiple layers of convolutional and pooling operations, CNNs can discern increasingly abstract features, such as edges, textures, and shapes, ultimately providing rich representations of the input data.

In practice, researchers and practitioners leverage pre-trained CNN models, such as VGG, ResNet, or Inception [16,53,54], which have been trained on large-scale image datasets like ImageNet. These models serve as feature extractors, where the activations from intermediate layers are used as feature vectors for input images. The works of Muhammad Usman S et.al [55] and Jana R and Mukherjee [11] deploy such techniques to generate automated features from a dataset of EEG signals. Further, several papers study the feature production using existing neural networks [9]. Transfer learning techniques allow fine-tuning these models on specific datasets, adapting them to perform tasks like classification, detection, or segmentation between pre-ictal, ictal and post-ictal episodes. This approach not only saves computational resources but also yields superior results compared to traditional handcrafted features, particularly in scenarios where data is high-dimensional or exhibits intricate structures. Thus, CNNs have become indispensable tools for generating rich and discriminative features, empowering various machine learning applications across domains.

Further, combining statistical and neural network generated features seem to have overtaken majority of the manual feature extraction approaches. The proposed method for feature extraction in the works of Usman et.al [55] involves both handcrafted and automated features. Handcrafted features include statistical and spectral moments, while automated features are extracted using a customized convolutional neural network (CNN). The CNN takes as input the 3 dimensional tensor of size m-by-n-by-c, that is formed from m-by-n feature matrices derived for each of the EEG channels [55]. Short Time Fourier Transform (STFT) is applied on a 30-second overlapping window with an overlap of 15 s for EEG signals. The CNN architecture consists of three layers, with 16 filters of 5 × 5 applied in the first layer, followed by 16 filters of 3 × 3 and 32 filters of 3 × 3 in the second and third layers, respectively. A flatten layer is used to extract the feature set. Both handcrafted and automated features are combined to form a feature vector.

The CNN architecture used in this study consists of three convolutional layers. In the first layer, 16 filters of size 5 × 5 are applied to the input spectrograms, where each filter performs a convolution operation across the input data, producing a set of feature maps that capture local spatial patterns. This convolution results in smaller matrices compared to the input spectrograms, reducing the dimensionality while retaining key features. The feature maps from the first layer are then processed in the second convolutional layer by 16 filters of size 3 × 3, which further extract more detailed features by focusing on smaller receptive fields. Finally, the third layer applies 32 filters of size 3 × 3 to the output of the second layer, enhancing the network's ability to capture intricate patterns in the data.

Regarding the multi-channel input data, each EEG channel is treated as a separate input, and the CNN applies the convolution filters across all channels simultaneously. This means that the convolution layers learn spatial patterns from the combined channels rather than processing each channel independently. This approach allows the network to capture relationships between channels, which can be important for detecting inter-channel correlations indicative of seizures. After the convolution layers, a flatten layer is used to transform the multidimensional feature maps into a single feature vector, combining both handcrafted features (such as statistical features from EEG signals) and automatically learned features from the CNN to form a comprehensive feature set. To minimize the curse of dimensionality, feature selection techniques such as Pearson Correlation Coefficient (PCC) and Particle Swarm Optimization (PSO) are applied to retain only the most relevant and distinguishing features from the combined feature set. This ensures the resulting feature vector remains compact yet informative, enhancing classification performance without redundancy [55].

Recently, newer deep learning architectures have also been introduced to improve the quality and efficiency of automated feature extraction from EEG signals. For instance: Temporal Convolutional Networks (TCN) have demonstrated superior ability in handling long-range temporal dependencies while maintaining a simple architecture. Their dilated convolution layers allow modeling over long time periods with fewer parameters, which is ideal for EEG time-series data [56]. Transformer-based models, such as the Vision Transformer (ViT) and EEG Transformer architectures, employ attention mechanisms to learn relationships across time steps without relying on recurrence. This allows for rich feature representations and better generalization across subjects [57]. LSTM-CNN Hybrid models leverage the strength of CNNs for spatial feature extraction and LSTM (Long Short-Term Memory) networks for modeling temporal dynamics. These hybrid architectures have proven effective in EEG-based seizure prediction due to their ability to jointly learn spatial-temporal patterns [58]. MAMBA (Multi-scale Adaptive Memory-Based Attention) is a recent innovation that integrates adaptive memory and attention mechanisms, enabling the model to capture dynamic temporal dependencies with reduced computational cost. Its potential in EEG modelling is under investigation, showing promising results in dynamic sequence modelling. These newer models not only extract richer features but also improve model robustness, reduce overfitting, and support real-time inference through architectural innovations. The field is rapidly advancing, and such architectures hold significant promise for next-generation seizure prediction systems.

Traditional handcrafted features and AI-derived features differ fundamentally in their design philosophy, computational demands, and intended application contexts. Handcrafted features, such as statistical moments, spectral band power, entropy measures, and wavelet coefficients, are explicitly designed using domain knowledge from signal processing and clinical neuroscience. Their primary advantage lies in their high interpretability and low computational cost, making them well suited for real-time and resource-constrained seizure prediction systems, particularly in patient-specific settings where available training data are limited. Because these features are mathematically well defined and clinically meaningful, they also facilitate easier validation, debugging, and clinical acceptance. However, handcrafted features may struggle to capture complex non-linear and cross-channel interactions present in EEG data, which can limit their generalization performance across heterogeneous patient populations.

In contrast, AI-derived features are automatically learned by deep learning models such as convolutional neural networks, transformers, and temporal convolutional networks. These features excel at modeling complex spatial-temporal dependencies and non-linear dynamics in high-dimensional EEG data, often resulting in superior performance for generalized, multi-patient seizure prediction tasks. Their robustness to noise and ability to learn hierarchical representations make them particularly effective when large, diverse datasets are available. Nevertheless, AI-derived features typically require higher computational resources, larger training datasets, and additional optimization for real-time or edge deployment. Furthermore, their reduced interpretability can hinder clinical trust and limit their direct applicability in personalized scenarios with scarce data. Consequently, recent research trends favor hybrid frameworks that combine the interpretability and efficiency of handcrafted features with the representational power of AI-derived features, achieving a balanced trade-off between accuracy, personalization, and computational efficiency.

Table 4 summarizes the key trade-offs between handcrafted and AI-derived features, illustrating that neither approach is universally superior; rather, their effectiveness depends on the prediction objective, data availability, and deployment constraints.

3.3. Classification

Several machine learning techniques have been widely applied to the task of epileptic seizure prediction, leveraging biomedical signals like electroencephalography (EEG) to forecast seizure occurrences. Support Vector Machines (SVMs) have been particularly popular due to their ability to handle high-dimensional data and their effectiveness in binary classification tasks. SVMs create hyperplanes that best separate seizure and non-seizure data based on features extracted from EEG signals. For example, Shoeb and Guttag [59] demonstrated the capability of SVMs in predicting seizures, achieving high accuracy by using temporal and spectral features of EEG signals [60].

In addition to SVMs, decision tree-based models like Random Forests have shown promise for seizure prediction. Random Forests combine multiple decision trees to improve predictive performance, making them more robust against overfitting, particularly in noisy or complex datasets. Studies like Mursalin et al. [60] have employed Random Forest classifiers, reporting notable success in distinguishing between pre-ictal (pre-seizure) and inter-ictal (non-seizure) states using a variety of EEG-derived features [60].

Neural Networks, including both traditional feedforward networks

Table 4
Comparison of Handcrafted and AI-Derived Features in Seizure Prediction.

Aspect	Handcrafted features	Ai-derived features
Best suited for	Personalized, patient-specific prediction	Generalized, multi-patient prediction
Data requirements	Low; effective with limited data	High; benefits from large datasets
Computational cost	Low; suitable for real-time and edge devices	High; often requires gpus and optimization
Interpretability	High; clinically and mathematically interpretable	Low; black-box representations
Modeling capability	Limited non-linear and cross-channel interactions	Strong spatial–temporal modeling
Clinical applicability	Easier validation and clinical acceptance	Higher accuracy but limited transparency

and more advanced deep learning models, have also been utilized for seizure prediction. Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) are particularly effective for modeling temporal dependencies and extracting spatial patterns from EEG data [61]. Ictal states are often characterized by subtle, non-linear patterns that these networks can capture. For instance, Truong et al. [56] explored CNNs for seizure prediction, reporting improved results in patient-specific prediction models. These neural networks excel at automatic feature extraction in addition to classification, allowing for more sophisticated temporal analysis compared to traditional machine learning techniques. The application of decision trees, SVMs, Random Forests, and Neural Networks offers diverse advantages in terms of interpretability, accuracy, and computational efficiency. However, neural networks, particularly deep learning models, are often more computationally expensive, prompting recent efforts to optimize these models for real-time applications and sustainability concerns

3.4. Performance metrics

Evaluating the performance of epilepsy seizure prediction models involves using specific metrics to assess their effectiveness. The choice of metrics depends on the nature of the problem and the characteristics of the data. Here are some commonly used performance metrics in the context of epilepsy seizure prediction:

- Sensitivity (True Positive Rate, Recall): The percentage of actual seizures correctly predicted by the model.

$$Sensitivity = TruePositive \ / \ (TruePositive + FalseNegative) \tag{1}$$

- Specificity (True Negative Rate):The percentage of non-seizure instances correctly identified as non-seizures.

$$Specificity = TrueNegative \ / \ (TrueNegative + FalsePositive) \tag{2}$$

- Accuracy:The overall correctness of the model, measuring the proportion of correctly predicted instances.

$$Accuracy = (TruePositive + TrueNegative) \ / \ (TotalSamples) \tag{3}$$

- Precision (Positive Predictive Value): The percentage of predicted seizures that are actually seizures.

$$Precision = TruePositive / \ (TruePositive + FalsePositive) \tag{4}$$

- False Positive Rate (FPR): The percentage of non-seizure instances incorrectly classified as seizures.

$$FPR = FalsePositive \ /(FalsePositive + TrueNegative) \tag{5}$$

- False Negative Rate (FNR): The percentage of actual seizures incorrectly classified as non-seizures.

$$FNR = FalseNegative/(FalseNegative + TruePositive) \tag{6}$$

- Detection / Prediction Time: The detection or prediction time refers to the time-period between occurrence of the seizure and first detection of seizure via the developed classifier.
- Successful Prediction: In the context of seizure prediction, a “successful” prediction is typically defined as the correct identification of a pre-ictal state within a clinically meaningful prediction horizon prior to seizure onset, while maintaining an acceptable false alarm rate. However, the definition of success varies across studies depending on prediction horizon length, seizure occurrence period, evaluation metrics, and clinical constraints.

Table 5 summarizes how each performance metric translates to clinical seizure prediction, their practical priority, and the consequences of low scores. These performance metrics, encapsulate various dimensions of the study outcome, with each metric holding specific relevance to the researcher's goals. The detailed goals of research often require a specific focus on certain metrics, leading to a varying order of importance in different scholarly works.

In the specialized domain of real-time seizure detection and prediction literature, a discernible trend emerges, where a predominant emphasis is placed on specific key metrics. Sensitivity, as a crucial metric, gauges the effectiveness of the system in accurately identifying seizures, thereby reflecting its sensitivity to true positive events. False positive predictions remain a critical challenge in seizure prediction systems and can significantly impact patient quality of life. Frequent false alarms may lead to increased anxiety, loss of trust in the system, alarm fatigue among caregivers, and unnecessary behavioral or clinical interventions. In real-world settings, the cost of false positives may therefore outweigh marginal gains in sensitivity. This highlights the importance of balanced evaluation metrics, patient-centered thresholds, and context-aware alarm strategies when designing and deploying seizure prediction models. Future research should place greater emphasis on minimizing false positives while maintaining clinically

Table 5
Clinical interpretation and importance of common performance metrics in epileptic seizure detection and prediction systems.

Metric	Description	High Values	Low Values	Medical priority
Sensitivity (Recall, TPR)	% of true seizures correctly predicted	Most seizures are caught; patient safety maximized	Dangerous under-detection; seizures missed	Maximized
False-Negative Rate (FNR)	% of seizures the classifier misses	Many seizures missed; no warning, risk of injury	Near-zero missed events	Minimized
Prediction Time	Minutes between onset and alert	Timely warnings allow intervention	Alerts too late to act	Earlier is always better, but not at the expense of sensitivity
False-Positive Rate / h (FPR)	Non-seizure segments flagged per hour	Frequent false alarms; alarm fatigue	Few unnecessary alarms; higher usability	Minimized, but secondary to not missing seizures
Precision (PPV)	Fraction of alarms that are real seizures	Few false alarms; good user trust	Many false alarms; users may ignore system	High precision helps usability but not as critical as catching seizures
Specificity (TNR)	% of non-seizure periods correctly ignored	System rarely produces an alarm for non-seizure	More false positives	Matters for comfort & workload; lower clinical risk than low sensitivity
Accuracy	Overall % correct decisions	Indicates balanced performance when classes are balanced	Can be misleading if seizures are rare	Not relied upon alone because class imbalance is huge
F1 Score	Harmonic mean of Precision & Recall	Balanced catch vs. false alarm when classes balanced	Harder to interpret in highly imbalanced data	Rarely reported in EEG studies; sensitivity & FPR/h are preferred

acceptable sensitivity levels. Additionally, the temporal aspects of prediction or detection time are prioritized, acknowledging the significance of prompt and timely responses in clinical settings [11,16,57,58].

This discerning focus on sensitivity, False Positive Rate per Hour, and prediction or detection time underscores the shared recognition within the academic community of their pivotal roles in evaluating the efficacy and practical utility of real-time seizure detection and prediction systems. As research endeavours continue to evolve, this emphasis on specific performance metrics serves as a guiding framework, ensuring a comprehensive understanding and evaluation of the advancements in this critical field of study.

The importance of efficiency in the development and deployment of AI models cannot be overstated in the face of growing environmental concerns [9,11,62]. AI models, particularly those powered by large-scale computations and deep learning, often require significant computational resources. The energy consumption associated with training and running these models contributes to carbon emissions and, consequently, exacerbates the environmental impact. As the demand for AI continues to rise, addressing the carbon footprint of AI models becomes imperative to align technological advancements with global efficiency goals.

Numerous studies have shed light on the environmental implications of AI, emphasizing the need for efficient practices in AI development. A research paper by Strubell et al. [63] highlights the substantial carbon footprint of training large language models, bringing attention to the environmental consequences of unchecked AI expansion. Considering this, efforts to enhance the energy efficiency of AI algorithms and infrastructure, as well as exploring renewable energy sources for data centers, are critical steps towards creating a more efficient AI ecosystem. Integrating sustainability considerations into AI model design and development is pivotal for ensuring that technological progress aligns with ecological responsibility. Therefore, this review paper aims to closely investigate the impacts of AI models, specifically in the field of seizure detection, in terms of efficiency and carbon footprint. This investigation is presented in Section 5, which evaluates optimization strategies such as input reduction, model simplification, and hardware-aware implementation to address these concerns.

4. Generalized vs. personalized models for seizure prediction

Epileptic seizure prediction has garnered significant attention in recent literature, with researchers exploring various approaches to enhance the accuracy and applicability of predictive models. One pivotal consideration in model development is the choice between generalized models, designed for broad populations, and patient-specific models, tailored to individual cases [42,64–67].

Generalized models, as proposed by [16,66] aim to capture universal patterns in seizure onset across diverse patient cohorts. By leveraging large datasets encompassing a spectrum of epilepsy cases, these models seek to identify common features that transcend individual differences. The study conducted by [16,66] demonstrated the efficacy of generalized models in identifying broad trends and commonalities in seizure patterns.

In contrast, patient-specific models, advocated by [42,67] recognize the heterogeneity of epilepsy manifestations and emphasize the need for personalized approaches. Drawing on comprehensive individualized data, including EEG recordings and medical histories, these models aim to capture the idiosyncrasies of each patient's seizures. The work of [42,65] showcases the potential of patient-specific models in achieving higher accuracy by accounting for individual variations.

Several factors influence the choice between these models. Data scarcity remains a fundamental challenge in personalized seizure prediction models, as individual patients often provide limited labeled seizure data for training and validation. This constraint increases the risk of overfitting, leading to unstable model performance and reduced generalizability, particularly for data-intensive deep learning architectures. As a result, many personalized models rely on simplified architectures, transfer learning, or adaptation from generalized models to mitigate limited data availability. Addressing data scarcity through robust personalization strategies, data-efficient learning, and cross-patient knowledge transfer remains an open and critical research challenge. Additionally, Dissanyake et al. [65] discuss the challenge of transferability, noting the potential limitations of patient-specific models in cases with limited data. The summary of personalized and generalized approaches is detailed in Table 6.

A promising avenue, as suggested by Afsari et al. [3] involves hybrid approaches that combines the strengths of both generalized and patient-specific models. By initially training on a broader dataset and subsequently fine-tuning with patient-specific information, these models aim to strike a balance between generalizability and individualization.

Beyond the inherent intricacies of the model, it becomes evident that handling a larger volume of data necessitates increased pre-processing and training time. Consequently, generalized models exhibit a substantial surge in energy consumption when compared to their individual counterparts [16]. It is noteworthy that in the development of these generalized models, crucial patient-specific details like age, gender, and seizure type are often overlooked in the predictor's design [3,55,68]. While concentrating solely on EEG data during the ictal and pre-ictal states yields relatively accurate models, the resultant output may be confined to a discrete binary classification indicating the presence or absence of a seizure. In contrast, incorporating personalized patient information such as age, gender, or vigilance presents a promising avenue to generate more informative responses, providing nuanced insights into the ongoing ictal or pre-ictal states. This approach allows for a more sophisticated understanding of the patient's condition, surpassing the limitations of a simple binary classification.

Generalizable models in healthcare aim to learn from data of individual patients and then fine-tune with new patient data to enhance performance and applicability. Initially, these models capture specific patient characteristics, and as they are exposed to more diverse data, they adapt and improve. This method benefits from high personalization and adaptability, making models efficient and effective in various medical scenarios. Foundation models, for instance, provide a robust framework for adapting to new tasks with fewer labeled examples, significantly improving the efficiency and scalability of medical AI systems [76].

Adaptability of classifiers seem to be the potential solution to the trade-off between the two types of models. The paper by Di Wu, Jie Yang, and Mohamad Sawan [77] introduces a novel training scheme that leverages knowledge distillation to enhance seizure prediction performance by utilizing patient-specific data. The authors demonstrate the effectiveness of their approach by applying it to five state-of-the-art baselines on an open-source dataset, consistently improving prediction accuracy. The literature review in the paper emphasizes the role of knowledge distillation in optimizing neural networks for healthcare applications and highlights the importance of avoiding the transfer of erroneous knowledge from teacher to student models. Overall, the research contributes to bridging the gap between patient-specific and patient-independent training methods in neurological disorder applications, showcasing the potential of personalized data and knowledge distillation techniques in improving predictive models for healthcare.

Further, the research carried out by Dissanayake et al. [65] signifies the utilization of transfer learning methods for patients that lack sufficient amount of data. The findings of the research indicate that presence of all patients in the dataset improves the accuracy of the model significantly while generalizing a model on the basis of limited patients is still a challenging task [65].

In conclusion, the choice between generalized and patient-specific models heavily depends on factors such as data availability, the intended scope of application, and the trade-off between broad applicability and personalized precision. The ongoing research in the field, shown by

Table 6
Comparative analysis of generalized and patient-specific seizure prediction models across recent literature.
This table summarizes representative studies, categorized by model type (generalized vs. patient-specific), dataset used, number of patients included in the study, reported sensitivity, prediction time, and false positive rate per hour (FP/H). A value of "1″ in the patient column indicates a single-patient (patient-specific) study rather than the first patient in a cohort. The comparison highlights the performance differences and practical trade-offs between generalized and individualized seizure prediction approaches.

Authors	Dependency	Data	Number of patients	Sensitivity	Prediction time	FP/H
Ilyas et al. (2023) [18]	Patient-independent	Own	15	92	40 min	NA
Assali et al. (2023) [17]	Patient-independent	CHB-MIT	23	92.8 – 88.6	30,60 min	NA
Sarvi Zargar et al. (2023) [16]	Patient-independent	European data base	7	98.47	40 min	0.031
Sarvi Zargar et al. (2023) [16]	Patient-specific	European data base	1	98.39	40 min	0.029
Khan et al. (2018) [69]	Patient-specific	CHB-MIT	1	87.8	10 min	0.142
Detti et al. (2019) [70]	Patient-specific	CHB-MIT	1	97.99	2 min	NA
Xu et al. (2023) [71]	Patient-specific	CHB-MIT	1	90.53	30 min	0.107
Massoud et al.(2023) [72]	Patient-specific	Melbourne University	1	70	10 min	0.41
Tariq et al. (2022) [73]	Patient-specific	CHB-MIT	1	93.8	NA	0.09
Rukhsar et al. (2019) [74]	Patient-specific	CHB-MIT	1	88.83	60 min	0.39
Dissanayake et al. (2017) [65]	Patient-independent	CHB-MIT	24	93.45	60 min	NA
Billeci et al. (2018) [64]	Patient-specific	PlosONE	15	89.06	15 min	0.41
Zhang et al. (2015) [75]	Patient-specific	American Epilepsy Society Seizure Prediction Challenge database	5 (dogs)	100	10 min	NA

the studies referenced, implies the importance of a novel approach to optimize the effectiveness of epileptic seizure prediction models.

5. Optimization

Given that the concept of sustainability in AI is relatively new, establishing a universal measure for it is challenging. While electrical power consumption is often considered, many factors influence this metric. For instance, running the same classifier on different types of hardware will lead to varying power consumption levels. The literature in this field often uses one the following methods to indicate incorporation of efficiency concepts in AI models:

(a) Input Reduction

Reducing the volume of data input in AI models can significantly lower power consumption, enhancing the overall efficiency of these systems. By minimizing the amount of data that needs to be processed, the computational load decreases, leading to fewer required operations and thus less energy usage [63]. Techniques such as data pruning, dimensionality reduction, and efficient data sampling can help achieve this reduction without compromising the model's accuracy. For instance, by focusing on the most relevant features or employing techniques like Principal Component Analysis (PCA), the data can be made more manageable and less resource-intensive to process. This not only conserves energy but also speeds up the processing time, making the models more efficient and environmentally friendly. Consequently, optimizing data input size is a crucial strategy in the pursuit of efficient AI practices [63].

(b) Model Optimization

Optimizing the structure of AI models plays a crucial role in reducing power consumption and enhancing the overall efficiency of these systems [78]. Model structure optimization involves refining the architecture of the neural network to achieve the desired performance with minimal computational resources. Techniques such as pruning, quantization, and the development of lightweight models like MobileNets and EfficientNet are commonly used.

These optimizations not only cut down on the energy required for training and inference but also enable the deployment of AI models on edge devices with limited computational power and battery life. Consequently, model structure optimization is a key strategy in developing efficient AI systems that are both powerful and energy-efficient [78–80].

(c) Application Specific Hardware/offloading to cloud

In particular cases, generic processors and GPUs may not be the best hardware to process the information. The works of Elhosary, et al. , Wei, et al. , Saric , et al. [81–83] show great potential in tailoring a hardware specifically for certain AI tasks such feature extraction and classification. Although a drawback of this approach is the lack of adaptability of the model.

The records in Table 7 and Fig. 7, are collected from a number of recent papers that specifically discuss lightweight, real-time, optimized solutions for seizure prediction. The research indicates the work in this area is focused on input reduction by reducing the number of EEG channels used for the classifiers. Further, hardware implementations specifically utilize reconfigurable hardware pieces such as FPGAs.

Wearable and edge-based technologies play an increasingly important role in enabling real-time seizure monitoring and prediction. Deploying prediction models on edge devices allows continuous biosignal acquisition and on-device inference with reduced latency and improved responsiveness. Wearable platforms further support personalized and real-time monitoring by facilitating long-term data collection in naturalistic settings. However, challenges related to sensor reliability, power consumption, and computational constraints remain key considerations for integrating wearable technology into practical seizure prediction systems.

6. Discussion and research gaps

Despite strong advancements in seizure prediction research, several limitations and gaps remain, particularly concerning personalized models and the integration of patient-specific features such as gender, age, and seizure type. Additionally, there are inherent limitations in the predictive capabilities of these models in terms of prediction time and resources, necessitating further optimization efforts to increase the overall efficiency of the model.

Table 7
Heatmap of optimization strategies in recent seizure prediction studies. A tick indicates the use of input reduction, model optimization, or hardware-specific implementation in each study.

Author	Input Reduction	Model Optimization	Application Specific Hardware
Olokodana et al. (2021) [84]			✓
Jana et al. (2021) [11]	✓		
Birjandtalab et al. (2017) [85]	✓		
Sayeed et al. (2018) [86]	✓		
Jose et al. (2020) [87]			✓
Saric et al. (2020) [82]			✓
Sahani et al. (2021) [88]		✓	✓
Idrees et al. (2022) [89]	✓		
Elhosary et al. (2021) [81]	✓		✓
Burrello et al. (2021) [90]	✓		
Rafiammal et al. (2020) [91]	✓		✓
Beeraka et al. (2022) [92]			✓
Dan et al. (2020) [93]		✓	
Qiu et al. (2022) [94]	✓		✓
Jana et al. (2023) [95]	✓		
Khalkhali et al. (2021) [53]	✓		
Chakrabarti et al. (2020) [96]			✓

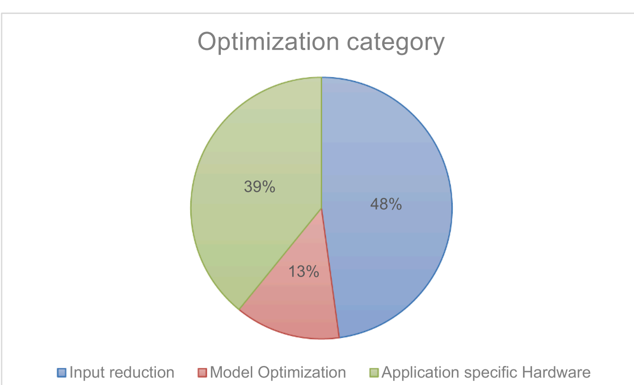

Fig. 7. Distribution of Optimization Strategies in Recent Seizure Prediction Research.

6.1. *Challenges in patient-centric personalization and data heterogeneity*

Personalized models for seizure prediction are designed to account for the unique characteristics of individual patients, theoretically offering superior accuracy compared to generalized models. However, these models face several challenges:

1. Data Availability and Quality: Personalized models require extensive and high-quality data from individual patients, including long-term EEG recordings and detailed medical histories. Obtaining such comprehensive data is often difficult due to privacy concerns, the need for continuous monitoring, and the logistical challenges of maintaining high-quality data over extended periods.
2. Heterogeneity in Epilepsy Manifestations: Epileptic seizures vary widely among patients in terms of frequency, duration, and intensity. This heterogeneity makes it challenging to develop models that can accurately predict seizures for all individuals. Personalized models must be meticulously tailored to capture these individual differences, which requires significant computational resources and expertise.
3. Transferability and Generalization: While personalized models excel in individual-specific contexts, their application is limited when trying to generalize findings to a broader population. This limitation hinders the development of scalable solutions that can benefit a wider range of patients.
4. Patient information: Age influences neurological structure and function, which in turn affects seizure characteristics [97]. Pediatric and geriatric patients, in particular, may exhibit seizure patterns that differ from those observed in adults. Incorporating age-specific information into prediction models is therefore important, yet it is often underrepresented in existing work [8,98]. In practice, age has been integrated in prior studies either through stratified analyses, age-conditioned feature extraction, or implicitly via patient-specific training and trial-based personalization. Gender can also influence the presentation and progression of epilepsy, with hormonal differences affecting seizure susceptibility and dynamics [99]. However, gender is typically incorporated only indirectly through personalized or patient-specific models, rather than as an explicit modeling variable, which may introduce bias or limit generalization. Further, variations in seizure frequency across patients can significantly influence model training, validation, and reported performance. However, many publicly available datasets do not provide sufficiently detailed or consistent information on patient-level seizure frequency and temporal variability, limiting deeper analysis in this review. As more comprehensive patient-specific data become available, future studies will be better positioned to investigate how seizure frequency impacts model learning dynamics, validation strategies, and personalization effectiveness.

Epilepsy encompasses a wide range of seizure types, each characterized by distinct temporal, spectral, and spatial EEG patterns. Seizure type is therefore a critical factor influencing prediction performance. Models that do not account for seizure-type variability risk producing overly generalized predictions that may not transfer reliably across patients or seizure phenotypes. Trial-based and personalized modeling approaches partially address this issue by tailoring models to individual seizure profiles; however, fully seizure-type-aware prediction remains constrained by the availability of detailed and consistently labeled datasets. This limitation highlights the need for systematic integration of patient-specific factors such as age, gender, and seizure type within personalized prediction frameworks.

6.2. *Computational efficiency and training data constraints*

Optimizing seizure prediction models involves balancing accuracy, computational efficiency, and real-time applicability. Several challenges persist in this domain:

- Training size: Training AI systems is a computationally exhaustive phase that involves processing extensive datasets to optimize the model's parameters. This process demands significant computational power, often requiring high-performance GPUs or TPUs and distributed computing resources to handle the massive parallel computations [100]. Thus, it is crucial to develop a model that requires a relatively small dataset for fine tuning the existing models.
- Computational Complexity: Advanced models, particularly those leveraging deep learning and neural networks, are computationally intensive. The high computational demand can limit the feasibility of real-time deployment, especially on portable or wearable devices with limited processing power.

- Feature Engineering, Feature Selection and Dimensionality: The transformation of raw biomedical signals into informative feature representations for model inference can be computationally intensive, particularly in real-time seizure prediction systems. Operations such as time–frequency transformations, spectral analysis, and multiscale decomposition introduce additional processing latency and energy consumption, which must be carefully considered in deployment scenarios. Moreover, the inclusion of a large number of features increases model dimensionality, potentially leading to overfitting and reduced generalizability, as well as higher computational overhead during inference. Effective feature selection and dimensionality reduction techniques are therefore essential to balance predictive performance with real-time processing constraints and energy efficiency.

6.3. Near-Real-Time deployment challenges

6.3.1. Technical challenges

Computationally expensive classifiers are typically developed without constraints on time and memory, making them suitable primarily for post-seizure analysis to enhance our understanding of seizures. However, this study aims to develop a proactive seizure predictor designed to alert caregivers before the onset of an ictal episode. The proposed real-time seizure classifier processes live data, continuously analyzing each incoming sample batch to detect potential seizure activity. These classifiers must operate efficiently to provide immediate and accurate predictions, ensuring timely interventions and improving patient outcomes.

While machine learning and deep learning methods have demonstrated promising performance in seizure prediction, existing algorithms exhibit several limitations, as summarized in Table 6. Many traditional machine learning approaches rely heavily on handcrafted features and are sensitive to feature quality and preprocessing choices, which can limit robustness across datasets. Deep learning models, although capable of learning complex representations, often require large amounts of labeled data and substantial computational resources, making them vulnerable to overfitting and difficult to deploy in real-time or resource-constrained environments. Furthermore, both model categories frequently exhibit limited generalizability across patients due to inter-subject variability and dataset bias. These limitations highlight the need for data-efficient learning strategies, improved generalization, and deployment-aware model design in future seizure prediction research.

Further, Long-term sustainability is an emerging concern in the deployment of AI-driven healthcare systems, including seizure prediction models. Computationally intensive training and inference pipelines may result in increased energy consumption, particularly when models are continuously updated or deployed at scale. These considerations highlight the importance of energy-efficient model design, lightweight architectures, and optimization techniques such as model compression and quantization. Addressing sustainability alongside performance and clinical reliability is essential for the responsible deployment of AI-based seizure prediction systems.

6.3.2. Caregiver training and human-system interaction

Successful deployment of seizure prediction systems also depends on appropriate training and education of caregivers and clinical staff. Caregivers must be able to correctly interpret prediction outputs, confidence levels, and alert notifications in order to respond appropriately and avoid unnecessary interventions or alarm fatigue. Inadequate training may lead to misuse or overreliance on predictions, potentially compromising patient safety. Therefore, caregiver education should be considered an integral component of deployment, alongside technical validation and system optimization.

In addition to algorithmic performance, user-friendly interfaces play a critical role in the effective deployment of seizure prediction systems. Caregivers and patients must be able to easily interpret alerts, confidence levels, and system status to make informed decisions in real time. Intuitive visualization, clear alert prioritization, and customizable notification settings can help reduce cognitive burden and support trust in the system. Emphasizing usability in interface design is therefore essential to ensure that seizure prediction technologies are practical, accessible, and safely adopted in clinical and home settings.

Beyond technical considerations, seizure prediction systems may also influence patient psychology, particularly in terms of anxiety, stress, and dependence on predictive alerts. Frequent false alarms or uncertain predictions may exacerbate psychological burden rather than alleviate it. These considerations highlight the importance of cautious deployment, transparent communication of system limitations, and integration with appropriate clinical guidance.

Although this review focuses on technological solutions, the development of reliable seizure prediction systems depends on interdisciplinary collaboration. In particular, medical professionals are essential for data acquisition, annotation, and seizure type labeling, while clinical and psychological expertise supports appropriate interpretation and deployment of prediction outputs.

6.3.3. Model complexity-interpretability trade-offs for clinical adoption

Model complexity and interpretability represent an important trade-off in seizure prediction systems, particularly for clinical adoption. While deep learning models with high representational capacity often achieve superior predictive performance, their black-box nature can limit clinical trust and transparency. In contrast, simpler or feature-based models offer greater interpretability but may sacrifice accuracy or robustness. This trade-off is further influenced by deployment constraints, as highly complex models may also incur increased computational costs and latency. Therefore, the selection of an appropriate model architecture depends on the clinical objective, deployment setting, and required level of explainability. Recent research highlights the need to balance predictive performance with interpretability, particularly in personalized seizure prediction systems where clinician understanding and patient safety are paramount.

6.3.4. Regulatory challenges

Regulatory considerations are an essential component of deploying AI-based seizure prediction systems in clinical settings. Issues related to patient data privacy, model bias and fairness, robustness under real-world conditions, and cybersecurity must be carefully addressed to ensure safe and ethical use. Although detailed regulatory frameworks are beyond the scope of this review, these factors should be considered alongside technical performance during system design and deployment.

6.4. Inherent unpredictability and reliability concerns in seizure prediction

Two key limitations can be noted that impact the performance of seizure prediction models:

- Unpredictable Nature of Seizures: Despite capturing patterns and correlations, seizures can occur unpredictably due to various internal and external factors. No model can guarantee 100 % accuracy in prediction, and false positives/negatives remain a significant concern.
- Latency in Detection: Real-time prediction models must process data with minimal latency to provide timely warnings. However, processing delays and detection latencies can limit the effectiveness of these models, particularly in critical situations.

6.5. Proposed hybrid seizure prediction model

The generalized backbone, foundational to our framework, is envisioned as a deep convolutional neural network (CNN) or a hybrid CNN-Recurrent Neural Network (RNN) architecture, potentially inspired by models proven effective in time-series analysis or multimodal bio-signal processing [58,101,102]. This backbone will be pre-trained on large, publicly available, multi-patient EEG, ECG, and EMG datasets to extract robust, high-level features that capture common pre-ictal patterns and physiological variations across a broad patient cohort.

The lightweight personalization layer will then serve as an adaptive module, designed for efficient fine-tuning with limited individual patient data. This layer could take the form of a small, fully connected neural network, a personalized support vector machine (SVM), or even a shallow decision tree ensemble. Its role is to translate the generalized features from the backbone into patient-specific predictions, modulated by individual patient profiles.

Crucially, Dynamic Time Warping (DTW) will be employed to address the inherent variability in seizure morphology and timing across patients. DTW can be applied either as a pre-processing step to align time-series segments (e.g., pre-ictal and inter-ictal phases) from different patients or even different seizures within the same patient, thereby standardizing signal patterns before feature extraction [103]. This alignment capability significantly minimizes the need for extensive, precisely synchronized training data and can dramatically reduce the duration of the training phase for the personalization layer.

From a deployment perspective, computational efficiency is a critical consideration for seizure prediction systems intended for real-time or edge-based operation. Optimization techniques such as similarity-based input reduction can significantly decrease input dimensionality by filtering redundant or low-information segments prior to inference [85, 95]. In addition, model compression strategies, including quantization and lightweight architecture design, enable on-board processing with reduced memory footprint and energy consumption. These approaches support the practical deployment of AI-based seizure prediction models without substantially compromising prediction performance.

Sustained deployment of seizure prediction systems necessitates ongoing evaluation and adaptive refinement to preserve performance over time. EEG characteristics and seizure dynamics can change due to disease progression, medication adjustments, lifestyle variations, and sensor-related factors. Without regular reassessment, AI models, personalized models in particular, may experience performance degradation. Consequently, mechanisms for continuous or periodic performance monitoring, model updating, and validation are essential to maintain long-term reliability and clinical relevance. In this context, structured patient feedback, such as confirmation of false alarms or perceived warning usefulness, can provide valuable information for refining personalized models and adjusting decision thresholds. Future research should therefore focus on adaptive learning and post-deployment evaluation frameworks that balance robustness, safety, computational efficiency, and patient engagement.

To further ensure computational efficiency and facilitate real-time performance on edge devices, a combination of optimization techniques is integrated. Feature selection methods will identify the most discriminative and computationally inexpensive features. Dimensionality reduction techniques such as PCA and autoencoders will compact the feature space, reducing memory footprint and processing load. Finally, model compression techniques will be applied to the entire hybrid architecture. This includes pruning, where redundant connections or neurons are removed from the neural network weights, and quantization, which reduces the precision of model parameters (e.g., from 32-bit floating-point to 8-bit integers) without significant performance degradation. These optimizations are paramount for deployment on portable, battery-powered devices with limited computational and memory resources, ensuring minimal latency from data acquisition to prediction output.

Despite promising theoretical foundations, many hybrid AI approaches in seizure prediction remain at the conceptual or proof-of-concept stage, with relatively limited empirical validation in real-world or clinically relevant datasets. This limitation is consistent with broader observations in hybrid modeling research, where most reported studies focus on demonstrating methodological feasibility rather than extensive empirical evaluation or deployment outcomes, especially in healthcare contexts [104].

Consequently, there is a need for more empirical evidence, including prospective validation, larger cohort studies, and real-world testing, to establish the practical effectiveness and robustness of hybrid seizure prediction models.

To ensure a comprehensive and clinically relevant evaluation of seizure prediction systems, a standardized set of robust performance metrics is essential. Beyond traditional classification metrics like accuracy, sensitivity, and specificity, the efficacy of real-time predictive models must be assessed using:

- Prediction Horizon (pH): The time interval between a successful prediction and the actual seizure onset.
- Seizure Prediction Rate (SPR) or Sensitivity: The percentage of seizures correctly predicted.
- False Positive Rate (FPR) per hour: The rate at which the system issues false alarms, a crucial metric for patient quality of life.
- Pre-ictal Warning Time (PWT): The duration for which an alert is active before a seizure.
- Latency: The time taken for the system to process data and issue a prediction after a pre-ictal change is detectable.
- Computational metrics: Including inference time (ms), memory footprint (MB), and power consumption (mW) for edge deployment scenarios.

A balanced optimization across these metrics, rather than solely focusing on sensitivity, is critical for real-world clinical utility.

Building on the challenges and opportunities discussed in this review, future research on seizure prediction should follow a structured and progressive roadmap, as illustrated in Fig. 8. The roadmap outlines the transition from data acquisition and feature representation to model development, personalization, and real-world deployment. Key directions include improving data quality and diversity through longitudinal and multi-center datasets, systematically integrating patient-specific factors such as age, gender, and seizure type into personalized models, and developing data-efficient and adaptive learning strategies to address data scarcity and overfitting. As highlighted in Fig. 8, optimization for real-time operation and energy-aware deployment, followed by post-deployment monitoring and feedback-driven model adaptation, represents a critical final stage in translating seizure prediction systems from research prototypes to clinically viable solutions.

7. Conclusions and future work

In this review, the state of the art in epileptic seizure prediction was explored, with a focus on personalized models, optimized AI practices, and the integration of multimodal biomedical signals such as EEG, ECG, and EMG. The survey highlights that while generalized models offer broad applicability, patient-specific models consistently demonstrate higher accuracy by accounting for individual variability in age, gender, and seizure type. It was emphasized that real-time seizure prediction requires not only accuracy but also timing efficiency and computational feasibility, especially for deployment in clinical or wearable settings. The resource implications of deep learning architectures were further examined, stressing the need for optimization strategies, including input reduction, model simplification, and the use of edge-computing or application-specific hardware. These are crucial for mitigating the energy footprint of AI models and ensuring sustainability. However, several challenges persist in current approaches. Personalized models are often constrained by data scarcity, particularly for patients with infrequent recorded seizures. Moreover, current models frequently overlook critical patient profile data, despite evidence that such information can significantly enhance prediction performance. Furthermore, many existing AI-based predictors still lack real-time robustness,

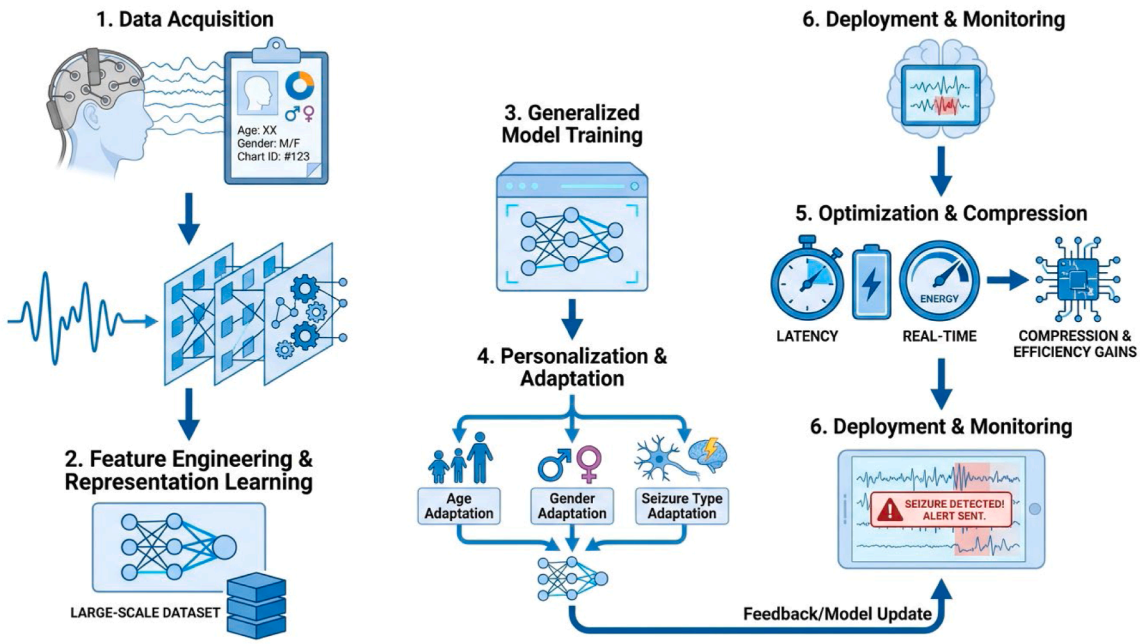

Fig. 8. Roadmap for Future Seizure Prediction Research and Deployment.

sufficient generalizability, or the ability to operate effectively across diverse clinical environments. To address these limitations, a novel hybrid prediction framework is proposed, integrating a generalized backbone model with a lightweight personalization layer. This framework aims to mitigate data scarcity by leveraging transfer learning from large, multi-patient datasets, while a patient-specific module fine-tunes predictions using limited individual data and clinical profiles. This approach achieves a critical balance between generalization and personalization, supporting efficient, sustainable deployment on edge devices, and enhancing interpretability. The proposed hybrid model offers a promising roadmap for future research, bridging key gaps in scalability, personalization, and computational efficiency, and ultimately advancing the real-world adoption of effective seizure prediction systems.

CRediT authorship contribution statement

Kiyan Afsari: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **May El Barachi:** Supervision. **Christian Ritz:** Supervision. **Abigail Copiaco:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] A. Copiaco, C. Ritz, S. Fasciani, N. Abdulaziz, An application for dementia patient monitoring with sound level assessment tool. In Proceedings of the 2020 3rd International Conference on Signal Processing and Information Security (ICSPIS), IEEE, DUBAI, United Arab Emirates, 2020, pp. 1–4.
[2] S.W. Muir, A.M. Yohannes, The impact of cognitive impairment on rehabilitation outcomes in elderly patients admitted with a femoral neck fracture: a systematic review, J. Geriatr. Phys. Ther. 32 (2009) 24–32, https://doi.org/10.1519/00139143-200932010-00006.
[3] K. Afsari, M. El Barachi, S. Fasciani, F. Belqasmi, A deep learning approach for real-time detection of epileptic seizures using EEG. In Proceedings of the 2022 7th International Conference on Smart and Sustainable Technologies (SpliTech), IEEE, Split /Bol, Croatia, 2022, pp. 1–7.
[4] F. Es.haghi, J. Frounchi, P. Shahabi, M. Sadighi, Absence epilepsy seizure onsets detection based on ECG signal analysis, in: Proceedings of the 2013 20th Iranian Conference on Biomedical Engineering (ICBME), IEEE Computer Society, 2013, pp. 219–222.
[5] Y. Guo, others epileptic seizure detection by cascading Isolation forest-based anomaly screening and EasyEnsemble, IEEE Trans. Neural Syst. Rehabil. Eng. 30 (2022) 915–924, https://doi.org/10.1109/TNSRE.2022.3163503.
[6] A. Shoeibi, M. Khodatars, N. Ghassemi, M. Jafari, P. Moridian, R. Alizadehsani, M. Panahiazar, F. Khozeimeh, A. Zare, H. Hosseini-Nejad, et al., Epileptic seizures detection using deep learning techniques: a review, Int J Env. Res Public Health 18 (2021) 5780, https://doi.org/10.3390/ijerph18115780.
[7] Epilepsy Available online: https://www.who.int/news-room/fact-sheets/detail/epilepsy (accessed on 9 April 2025).
[8] C.E. Stafstrom, L. Carmant, Seizures and epilepsy: an overview for neuroscientists, Cold. Spring. Harb. Perspect. Biol. 7 (2015) 1–19, https://doi.org/10.1101/cshperspect.a022426.
[9] H. Daoud, M.A. Bayoumi, Efficient epileptic seizure prediction based on deep learning, IEEe Trans. Biomed. Circuits. Syst. 13 (2019) 804–813, https://doi.org/10.1109/TBCAS.2019.2929053.
[10] S.M. Usman, S. Khalid, Z. Bashir, Epileptic seizure prediction using scalp electroencephalogram signals, Biocybern. Biomed. Eng. 41 (2021) 211–220, https://doi.org/10.1016/j.bbe.2021.01.001.
[11] R. Jana, I. Mukherjee, Deep learning based efficient epileptic seizure prediction with EEG channel optimization, Biomed. Signal. Process. Control 68 (2021) 102767, https://doi.org/10.1016/j.bspc.2021.102767.
[12] Jones, L.A.; Thomas, R.H. Sudden death in epilepsy: insights from the last 25 years. 2017, doi:10.1016/j.seizure.2016.10.002.
[13] R.S. Fisher, others instruction manual for the ILAE 2017 operational classification of seizure types, Epilepsia 58 (2017) 531–542, https://doi.org/10.1111/epi.13671.
[14] E. Trinka, A definition and classification of status epilepticus - report of the ILAE task force on classification of status epilepticus, Epilepsia 56 (2015) 1515–1523, https://doi.org/10.1111/epi.13121.
[15] SEIZURE TYPES & FIRST AID, Edmonton epilepsy association, 2021.
[16] B. Sarvi Zargar, M.R. Karami Mollaei, F. Ebrahimi, J. Rasekhi, Generalizable epileptic seizures prediction based on deep transfer learning, Cogn. Neurodyn. 17 (2023) 119–131, https://doi.org/10.1007/s11571-022-09809-y.
[17] I. Assali, A. Ghazi Blaiech, A. Ben Abdallah, K. Ben Khalifa, M. Carrère, M. Hédi Bedoui, CNN-based classification of epileptic states for seizure prediction using combined temporal and spectral features, Biomed. Signal. Process. Control 82 (2023) 104519, https://doi.org/10.1016/j.bspc.2022.104519.
[18] A. Ilyas, O.A. Alamoudi, K.O. Riley, S. Pati, Pro-ictal State in Human temporal lobe epilepsy, NEJM Evid. 2 (2023), https://doi.org/10.1056/EVIDoa2200187.
[19] Acharya, U.R.; Hagiwara, Y.; Adeli, H. Automated seizure prediction. 2018, doi:10.1016/j.yebeh.2018.09.030.

[20] Natu, M.; Bachute, M.; Gite, S.; Kotecha, K.; Vidyarthi, A. Review on epileptic seizure prediction: machine learning and deep learning approaches. 2022, doi:10.1155/2022/7751263.
[21] Ramgopal, S.; others Seizure detection, Seizure prediction, and closed-loop warning systems in epilepsy. 2014, doi:10.1016/j.yebeh.2014.06.023.
[22] S. Shafiezadeh, G. Marco Duma, M. Pozza, A. Testolin, A systematic review of cross-patient approaches for EEG epileptic seizure prediction, J. Neural Eng. 21 (2024) 061004, https://doi.org/10.1088/1741-2552/ad9682.
[23] M.K. Siddiqui, R. Morales-Menendez, X. Huang, N. Hussain, A review of epileptic seizure detection using machine learning classifiers, Brain Inf. 7 (2020) 5, https://doi.org/10.1186/s40708-020-00105-1.
[24] Kim, T.; others Epileptic seizure detection and experimental treatment: a review. 2020, doi:10.3389/fneur.2020.00701.
[25] B.A. Moore, J.E. Barnett, Niedermeyer's electroencephalography: basic principles, clinical applications, and related fields. Case Studies in Clinical Psychological Science: Bridging the Gap from Science to Practice, 2017, pp. 1–7, https://doi.org/10.1093/MED/9780190228484.001.0001.
[26] S.J. Hussain, Epileptic seizure detection using wavelets and EMD, in: Proceedings of the Proceedings of the 4th International Conference on Biosignals, Images and Instrumentation (ICBSII), Institute of Electrical and Electronics Engineers Inc., 2018. November.
[27] Elger, C.E.; Hoppe, C. Diagnostic challenges in epilepsy: seizure under-reporting and seizure detection. 2018, doi:10.1016/S1474-4422(18)30038-3.
[28] Guerreiro, C.A.M. Epilepsy: is there hope? 2016, doi:10.4103/ijmr.IJMR_1051_16.
[29] Shih, J.J.; others Indications and methodology for video-electroencephalographic studies in the Epilepsy Monitoring Unit. 2018, doi:10.1111/epi.13938.
[30] J. Jeppesen, Seizure detection based on heart rate variability using a wearable electrocardiography device, Epilepsia 60 (2019) 2105–2113, https://doi.org/10.1111/epi.16343.
[31] P. Kligfield, Recommendations for the standardization and interpretation of the electrocardiogram: part I: the electrocardiogram and its technology: a scientific Statement from the American Heart Association Electrocardiography and Arrhythmias Committee, Council on Clinical Cardiology; the American College of Cardiology Foundation; and the Heart Rhythm Society: endorsed by the International Society for Computerized Electrocardiology, Circulation 115 (2007) 1306–1324, https://doi.org/10.1161/CIRCULATIONAHA.106.180200.
[32] F. Shaffer, R. McCraty, C.L. Zerr, A healthy heart is not a metronome: an integrative review of the heart's anatomy and heart rate variability, Front. Psychol. (2014) 5, https://doi.org/10.3389/FPSYG.2014.01040.
[33] M. J. Wan, A.A.H. Al-awlaqi, M. Li, M. O'Grady, X. Gu, J. Wang, N. Cao, Wearable IoT enabled real-time health monitoring system J. Wirel. Com. Netw. 2018 (2018) 298, https://doi.org/10.1186/s13638-018-1308-x.
[34] I. Hussain, S.J. Park, HealthSOS: real-time health monitoring system for stroke prognostics, IEEe Access. 8 (2020) 213574–213586, https://doi.org/10.1109/ACCESS.2020.3040437.
[35] Y. Xu, J. Yang, W. Ming, S. Wang, M.Shorter Sawan, Latency of real-time epileptic seizure detection via probabilistic prediction, Expert. Syst. Appl. 236 (2024) 121359, https://doi.org/10.1016/j.eswa.2023.121359.
[36] K. Afsari, M. El Barachi, C. Ritz, Near-real-time epileptic seizure detection with reduced EEG electrodes: a BiLSTM-wavelet approach on the EPILEPSIAE dataset, Brain Sci. 16 (2026) 119, https://doi.org/10.3390/brainsci16010119.
[37] P.S. De Oliveira, M.A.A. Lima, A.S. Cerqueira, C.A. Duque, D.D. Ferreira, Harmonic analysis based on scica at PCC of a grid-connected Micro Solar PV Power plant, in: In Proceedings of the Proceedings of International Conference on Harmonics and Quality of Power (ICHQP), 2018, pp. 1–6. JuneVol. 2018-May.
[38] L. Kuhlmann, K. Lehnertz, M.P. Richardson, B. Schelter, H.P. Zaveri, Seizure prediction — ready for a new era, Nat. Rev. Neurol. 14 (2018) 618–630, https://doi.org/10.1038/s41582-018-0055-2.
[39] F. Fürbass, P. Ossenblok, M. Hartmann, H. Perko, A.M. Skupch, G. Lindinger, L. Elezi, E. Pataraia, A.J. Colon, C. Baumgartner, et al., Prospective Multi-center study of an automatic online seizure detection system for epilepsy monitoring units, Clin. Neurophysiol. 126 (2015) 1124–1131, https://doi.org/10.1016/j.clinph.2014.09.023.
[40] R.G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, C.E. Elger, Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: dependence on recording region and brain State, Phys. Rev .Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top. 64 (2001) 8, https://doi.org/10.1103/PhysRevE.64.061907.
[41] Shoeb, A. CHB-MIT Scalp EEG Database 2010.
[42] A. Shoeb, H. Edwards, J. Connolly, B. Bourgeois, S. Ted Treves, J. Guttag, Patient-specific seizure onset detection, Epilepsy Behav. 5 (2004) 483–498, https://doi.org/10.1016/j.yebeh.2004.05.005.
[43] EEG Database — Seizure Prediction Project Freiburg, (2012).
[44] I. Obeid, J. Picone, The Temple University Hospital EEG data corpus, Front. Neurosci. 10 (2016) 195498, https://doi.org/10.3389/fnins.2016.00196.
[45] M. Ihle, EPILEPSIAE - A European epilepsy database, Comput. Methods Programs Biomed. 106 (2012) 127–138, https://doi.org/10.1016/j.cmpb.2010.08.011.
[46] J. Maheshwari, S.D. Joshi, T.K. Gandhi, Real-time automated epileptic seizure detection by analyzing time-varying high spatial frequency oscillations, IEEe Trans. Instrum. Meas. (2022) 71, https://doi.org/10.1109/TIM.2022.3152325.
[47] T. Kim, N.S. Artan, I.W. Selesnick, H.J. Chao, Seizure detection methods using a cascade architecture for real-time implantable devices, in: Proceedings of the Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), Institute of Electrical and Electronics Engineers Inc., 2013, pp. 1005–1008.
[48] T. Zhang, W. Chen, X. Chen, Identifying epileptic EEGs and congestive heart failure ECGs under Unified framework of Wavelet scattering Transform, bidirectional weighted (2D)2PCA and KELM, Biocybern. Biomed. Eng. 43 (2023) 279–297, https://doi.org/10.1016/j.bbe.2023.01.002.
[49] D. Gajic, Z. Djurovic, S. Di Gennaro, F. Gustafsson, Classification of EEG signals for detection of epileptic seizures based on wavelets and statistical pattern recognition, Biomed Eng (Singap.) (2014) 26, https://doi.org/10.4015/S1016237214500215.
[50] Z. Liu, G. Yao, Q. Zhang, J. Zhang, X. Zeng, Wavelet scattering transform for ECG beat classification, Comput. Math. Methods Med. 2020 (2020), https://doi.org/10.1155/2020/3215681.
[51] N. Ghassemi, A. Shoeibi, M. Rouhani, H. Hosseini-Nejad, Epileptic seizures detection in EEG signals using TQWT and Ensemble Learning, in: Proceedings of the 2019 9th International Conference on Computer and Knowledge Engineering (ICCKE), Institute of Electrical and Electronics Engineers Inc., 2019, pp. 403–408. October.
[52] N. Shawki, T. Elseify, T. Cap, V. Shah, I. Obeid, J. Picone, A deep learning-based real-time seizure detection system, in: Proceedings of the 2020 IEEE Signal Processing in Medicine and Biology Symposium (SPMB) - Proceedings, Institute of Electrical and Electronics Engineers Inc., 2020. December.
[53] V. Khalkhali, N. Shawki, V. Shah, M. Golmohammadi, I. Obeid, J. Picone, Low latency real-time seizure detection using transfer deep learning, in: Proceedings of the 2021 IEEE Signal Processing in Medicine and Biology Symposium (SPMB) - Proceedings, 2021.
[54] H.S. Nogay, H. Adeli, Detection of epileptic seizure using pretrained deep convolutional neural network and transfer learning, Eur. Neurol. 83 (2021) 602–614, https://doi.org/10.1159/000512985.
[55] S. Muhammad Usman, S. Khalid, S. Bashir, A deep learning based ensemble learning method for epileptic seizure prediction, Comput. Biol. Med. 136 (2021) 104710, https://doi.org/10.1016/j.compbiomed.2021.104710.
[56] N.D. Truong, Convolutional neural networks for seizure prediction using intracranial and scalp electroencephalogram, Neural Netw. 105 (2018) 104–111, https://doi.org/10.1016/J.NEUNET.2018.04.018.
[57] D. Liu, X. Dong, D. Bian, W. Zhou, Epileptic seizure prediction using attention augmented convolutional network, Int. J. Neur. Syst. 33 (2023) 2350054, https://doi.org/10.1142/S0129065723500545.
[58] X. Lu, A. Wen, L. Sun, H. Wang, Y. Guo, Y. Ren, An epileptic seizure prediction method based on CBAM-3D CNN-LSTM model, IEEe J. Transl. Eng. Health Med. 11 (2023) 417–423, https://doi.org/10.1109/JTEHM.2023.3290036.
[59] A.H. Shoeb, J. Guttag, Application of machine learning to epileptic seizure detection, in: Proceedings of the International Conference on Machine Learning, 2010.
[60] M. Mursalin, Y. Zhang, Y. Chen, N.V. Chawla, Automated epileptic seizure detection using improved correlation-based feature selection with random forest classifier, Neurocomputing 241 (2017) 204–214, https://doi.org/10.1016/J.NEUCOM.2017.02.053.
[61] K. Afsari, C. Ritz, M. ElBarachi, Deep learning for EEG seizure prediction: impact of feature engineering and prediction window, in: In Proceedings of the 2025 IEEE International Conference on Signals and Systems (ICSigSys), 2025, pp. 48–53. November.
[62] S. Zhao, J. Yang, M. Sawan, Energy-efficient neural network for epileptic seizure prediction, IEEE Trans. Biomed. Eng. 69 (2022) 401–411, https://doi.org/10.1109/TBME.2019.2929053.
[63] E. Strubell, A. Ganesh, A. McCallum, Energy and policy considerations for deep learning in NLP, in: Proceedings of the Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, Stroudsburg, PA, USA, 2019, pp. 3645–3650.
[64] L. Billeci, D. Marino, L. Insana, G. Vatti, M. Varanini, Patient-specific seizure prediction based on heart rate variability and recurrence quantification analysis, PLoS One 13 (2018) e0204339, https://doi.org/10.1371/journal.pone.0204339.
[65] Dissanayake, T.; Fernando, T.; Denman, S.; Sridharan, S.; Fookes, C. Patient-independent epileptic seizure prediction using deep learning models. 2020.
[66] G.M. Sigsgaard, Y. Gu, Improving the generalization of patient non-specific model for epileptic seizure detection, Biomed. Phys. Eng. Express. 10 (2024) 015010, https://doi.org/10.1088/2057-1976/ad097f.
[67] M. Winterhalder, Spatio-temporal patient-individual assessment of synchronization changes for epileptic seizure prediction, Clin. Neurophysiol. 117 (2006) 2399–2413, https://doi.org/10.1016/j.clinph.2006.07.312.
[68] Y. Li, H. Li, Z. Zhong, J. Zhou, A real-time cross-patient seizure detection algorithm based on adaptive template matching, in: Proceedings of the 2023 6th International Conference on Electronics Technology (ICET), Institute of Electrical and Electronics Engineers Inc., 2023, pp. 1122–1126.
[69] H. Khan, L. Marcuse, M. Fields, K. Swann, B. Yener, Focal onset seizure prediction using convolutional networks, IEEE Trans. Biomed. Eng. 65 (2018) 2109–2118, https://doi.org/10.1109/TBME.2017.2785401.
[70] P. Detti, G.Z.M. De Lara, R. Bruni, M. Pranzo, F. Sarnari, G. Vatti, A patient-specific approach for short-term epileptic seizures prediction through the analysis of EEG synchronization, IEEE Trans. Biomed. Eng. 66 (2019) 1494–1504, https://doi.org/10.1109/TBME.2018.2874716.
[71] X. Xu, Y. Zhang, R. Zhang, T. Xu, Patient-specific method for predicting epileptic seizures based on DRSN-GRU, Biomed. Signal. Process. Control 81 (2023) 104449, https://doi.org/10.1016/j.bspc.2022.104449.
[72] Y.M. Massoud, M. Abdelzaher, L. Kuhlmann, M.A. Abd El Ghany, General and patient-specific seizure classification using deep neural networks, Analog. Integr. Circuits. Signal. Process. 116 (2023) 205–220, https://doi.org/10.1007/s10470-023-02153-z.

[73] Z. Bin Tariq, A. Iyengar, L. Marcuse, H. Su, B. Yener, Patient-specific seizure prediction using single seizure electroencephalography recording, in: Proceedings of the Studies in Computational Intelligence, Springer Science and Business Media Deutschland GmbH, 2022, pp. 295–308.
[74] S. Rukhsar, Y.U. Khan, O. Farooq, M. Sarfraz, A.T. Khan, Patient-specific epileptic seizure prediction in long-term scalp EEG signal using multivariate statistical process control, IRBM 40 (2019) 320–331, https://doi.org/10.1016/j.irbm.2019.08.004.
[75] Z. Zhang, K.K. Parhi, Seizure prediction using polynomial SVM classification, in: Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), 2015, pp. 5748–5751, https://doi.org/10.1109/EMBC.2015.7319698, 2015-November.
[76] How foundation models can advance AI in healthcare, (2022).
[77] D. Wu, J. Yang, M. Sawan, Bridging the gap between patient-specific and patient-independent seizure prediction via knowledge distillation, J. Neural Eng. 19 (2022) 036035, https://doi.org/10.1088/1741-2552/ac73b3.
[78] Wijgård, B.; Eng, T. Power consumption when using AIModels on microcontrollers. 2022.
[79] S. Han, J. Pool, J. Tran, W.J. Dally, Learning both weights and connections for efficient neural networks, Adv. Neural Inf. Process. Syst. (2015) 1135–1143, 2015-January.
[80] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.C. Chen, MobileNetV2: inverted residuals and linear bottlenecks, in: Proceedings of the Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2018, pp. 4510–4520. January.
[81] H. Elhosary, M.H. Zakhari, M.A. Elgammal, M.A. Abd El Ghany, K.N. Salama, H. Mostafa, Low-power hardware implementation of a support vector machine training and classification for neural seizure detection, IEEe Trans. Biomed. Circuits. Syst. 13 (2019) 1324–1337, https://doi.org/10.1109/TBCAS.2019.2947044.
[82] R. Sarić, D. Jokić, N. Beganović, L.G. Pokvić, A. Badnjević, FPGA-based real-time epileptic seizure classification using artificial neural network, Biomed. Signal. Process. Control 62 (2020) 102106, https://doi.org/10.1016/J.BSPC.2020.102106.
[83] Y. Wei, A review of algorithm & hardware design for AI-based biomedical applications, IEEe Trans. Biomed. Circuits. Syst. 14 (2020) 145–163, https://doi.org/10.1109/TBCAS.2020.2974154.
[84] I.L. Olokodana, S.P. Mohanty, E. Kougianos, R.S.EZcap Sherratt, A novel wearable for real-time automated seizure detection from EEG signals, IEEE Trans. Consum. Electron. 67 (2021) 166–175, https://doi.org/10.1109/TCE.2021.3079399.
[85] J. Birjandtalab, M. Baran Pouyan, D. Cogan, M. Nourani, J. Harvey, Automated seizure detection using limited-channel EEG and non-linear dimension reduction, Comput. Biol. Med. 82 (2017) 49–58, https://doi.org/10.1016/J.COMPBIOMED.2017.01.011.
[86] M.A. Sayeed, S.P. Mohanty, E. Kougianos, V.P. Yanambaka, H. Zaveri, A robust and fast seizure detector for IoT edge, in: In Proceedings of the Proceedings - 2018 IEEE 4th International Symposium on Smart Electronic Systems (iSES), 2018, pp. 156–160. July.
[87] J.P. Jose, M. Sundaram, G. Jaffino, FPGA implementation of epileptic seizure detection using ELM classifier, in: In Proceedings of the 2020 6th International Conference on Bio Signals, Images, and Instrumentation (ICBSII), 2020. February.
[88] M. Sahani, S.K. Rout, P.K. Dash, FPGA implementation of epileptic seizure detection using semisupervised reduced deep convolutional neural network, Appl. Soft. Comput. 110 (2021) 107639, https://doi.org/10.1016/J.ASOC.2021.107639.
[89] A.K. Idrees, S.K. Idrees, R. Couturier, T. Ali-Yahiya, An edge-fog computing-enabled lossless EEG data compression with epileptic seizure detection in IoMT networks, IEEe Internet. Things. J. 9 (2022) 13327–13337, https://doi.org/10.1109/JIOT.2022.3143704.
[90] A. Burrello, S. Benatti, K. Schindler, L. Benini, A. Rahimi, An ensemble of hyperdimensional classifiers: hardware-friendly short-latency seizure detection with automatic iEEG electrode selection, IEEE J. Biomed. Health Inf. 25 (2021) 935–946, https://doi.org/10.1109/JBHI.2020.3022211.
[91] S.S. Rafiammal, D.N. Jamal, S.K. Mohideen, Reconfigurable hardware design for automatic epilepsy seizure detection using EEG signals, Eng. Tec*hnol. Appl. Sci. Res.* 10 (2020) 5803–5807, https://doi.org/10.48084/ETASR.3419.
[92] S.M. Beeraka, A. Kumar, M. Sameer, S. Ghosh, B. Gupta, Accuracy enhancement of epileptic seizure detection: a deep learning approach with hardware realization of STFT, Circuits. Syst. Signal. Process. 41 (2022) 461–484, https://doi.org/10.1007/S00034-021-01789-4/TABLES/5.
[93] J. Dan, B. Vandendriessche, W. Van Paesschen, D. Weckhuysen, A. Bertrand, Computationally-efficient algorithm for real-time absence seizure detection in wearable electroencephalography, Int. J. Neural Syst. (2020) 30, https://doi.org/10.1142/S0129065720500355.
[94] S. Qiu, W. Wang, H. Jiao, LightSeizureNet: a lightweight deep learning model for real-time epileptic seizure detection, IEEE J. Biomed. Health Inf. 27 (2023) 1845–1856, https://doi.org/10.1109/JBHI.2022.3223970.
[95] R. Jana, I. Mukherjee, Efficient seizure prediction and EEG channel selection based on multi-objective optimization, IEEe Access 11 (2023) 54112–54121, https://doi.org/10.1109/ACCESS.2023.3281450.
[96] S. Chakrabarti, A. Swetapadma, A. Ranjan, P.K. Pattnaik, Time domain implementation of pediatric epileptic seizure detection system for enhancing the performance of detection and easy monitoring of pediatric patients, Biomed. Signal. Process. Control 59 (2020) 101930, https://doi.org/10.1016/J.BSPC.2020.101930.
[97] Aging changes in the nervous system: MedlinePlus medical encyclopedia, 2022.
[98] Semrud-Clikeman, M.; Ellison, P.A.T. Introduction to child clinical neuropsychology. In Child Neuropsychology; 2009; pp. 3–23.
[99] N. Foldvary-Schaefer, C. Harden, A. Herzog, T. Falcone, Hormones and seizures, Cleve. Clin. J. Med. 71 (2004), https://doi.org/10.3949/ccjm.71.suppl_2.s11. S11–S18.
[100] The computing power needed to train AI is now rising seven times faster than ever before, MIT Technol. Rev. (2019).
[101] A.M. Abdelhameed, H.G. Daoud, M. Bayoumi, Deep convolutional bidirectional LSTM recurrent neural network for epileptic seizure detection, in: In Proceedings of the 2018 16th IEEE International New Circuits and Systems Conference (NEWCAS), 2018, pp. 139–143. June.
[102] G. Xu, T. Ren, Y. Chen, W. Che, A one-dimensional CNN-LSTM model for epileptic seizure recognition using EEG signal analysis, Front. Neurosci. (2020) 14, https://doi.org/10.3389/fnins.2020.578126.
[103] H. Lerogeron, R. Picot-Clémente, A. Rakotomamonjy, L. Heutte, Approximating dynamic time warping with a convolutional neural network on EEG data, Pattern. Recognit. Lett. 171 (2023) 162–169, https://doi.org/10.1016/j.patrec.2023.05.012.
[104] A. Ahmadi, M. Fakhimi, C. Magnusson, Hybrid modelling using simulation and machine learning in healthcare, Comput. Oper. Res. 185 (2026) 107278, https://doi.org/10.1016/j.cor.2025.107278.